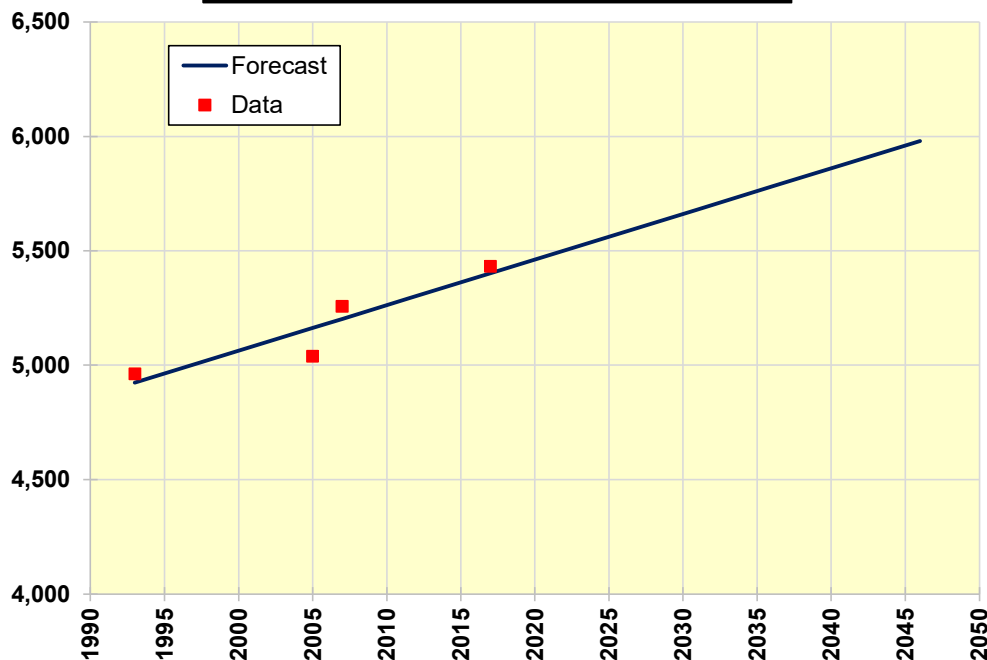
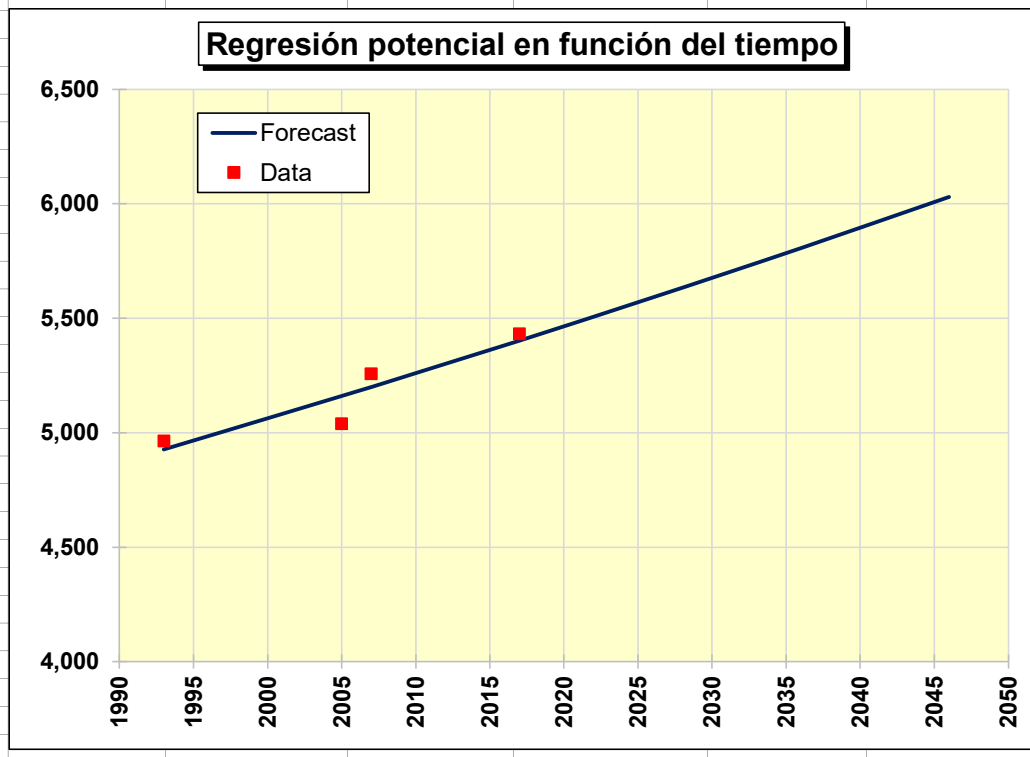


	A	B	C	D	E	F	G	H	I
1	LINEAL			IBM World-wide sales			INTERMEDIATE CALCULATIONS:		
2	Linear regression on time						RMSE (Square root of MSE)		71.99
3	$Y_i = a + bt$						3 X RMSE		215.97
4	Select Tools, Data Analysis, Regression. Enter ranges as follows:								
5	Y range includes column B, X range includes column C (mismo tamaño de la columna B), Output						Warm-up SSE		20730.85
6	Ensure number of data selected for Y and X ranges agree with the						Warm-up MSE		5182.71
7	nbr. of warm-up data entered in cell C10.						Forecasting SSE		20730.85
8							Fcst SSE - Warm-up SSE		0.00
9	INPUT:						Nbr. of forecast periods		0
10	Nbr. of warm-up data		4				Forecasting MSE		0.00
11	Last period to forecast		2046						
12									
13							Warm-up Sum abs. err.		247.58
14	OUTPUT:						Warm-up MAD		61.90
15	Number of data		4				Forecasting Sum abs. err.		247.58
16	Nbr. of outliers		0				Fcst Sum abs. - Warm-up Sum at		0.00
17	Warm-up MSE		5182.71				Nbr. of forecast periods		0
18	Forecasting MSE		0.00				Forecasting MAD		0.00
19	Warm-up MAD		61.90				Last forecast at period		2046
20	Forecasting MAD		0.00						
21									
22		Regression ranges							
23		Y range	X range						
24	Mon/yr.	Data	t	Est Y	e = Y - Est Y	e^2			
25	1993	4,963	1,993	4924	39.24	1539.82			
26	2005	5,039	2,005	5163	-123.79	15324.06			
27	2007	5,257	2,007	5203	54.37	2956.22			
28	2017	5,432	2,017	5402	30.18	910.75			
29	2026		2,026	5581.1	#N/D	#N/D			
30	2036		2,036	5780.3	#N/D	#N/D			
31	2046		2,046	5979.5	#N/D	#N/D			
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Regresión lineal en función del tiempo

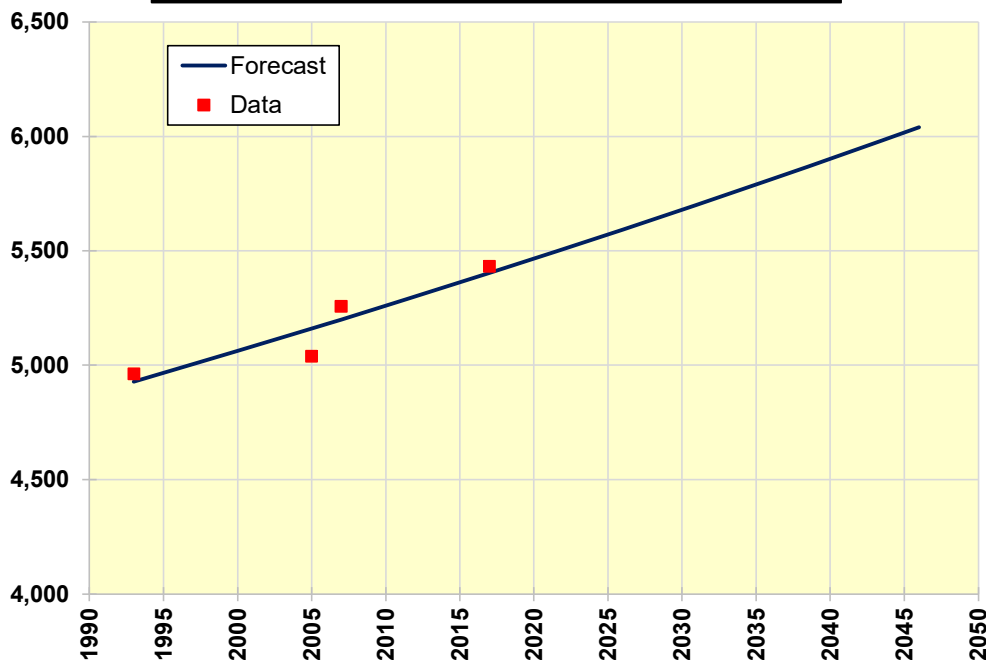


	A	B	C	D	E	F	G	H	I
1	POTENCIAL			IBM World-wide sales			INTERMEDIATE CALCULATIONS:		
2	Ponential regression on time						RMSE (Square root of MSE)		70.86
3	$\ln(Y) = a + b \cdot \ln(t)$						3 X RMSE		212.58
4	Select Tools, Data Analysis, Regression. Enter ranges as follows:						Warm-up SSE		20084.88
5	Y range includes column E, X range includes column D (mismo tamaño de la columna E), Output						Warm-up MSE		5021.22
6	Ensure number of data selected for Y and X ranges agree with the						Forecasting SSE		20084.88
7	nbr. of warm-up data entered in cell C10.						Fcst SSE - Warm-up SSE		0.00
8							Nbr. of forecast periods		0
9	INPUT:						Forecasting MSE		0.00
10	Nbr. of warm-up data		4				Warm-up Sum abs. err.		243.90
11	Last period to forecast		2046				Warm-up MAD		60.97
12							Forecasting Sum abs. err.		243.90
13							Fcst Sum abs. - Warm-up Sum at		0.00
14	OUTPUT:						Nbr. of forecast periods		0
15	Number of data		4				Forecasting MAD		0.00
16	Nbr. of outliers		0				Last forecast at period		2046
17	Warm-up MSE		5021.22						
18	Forecasting MSE		0.00						
19	Warm-up MAD		60.97						
20	Forecasting MAD		0.00						
21									
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23									
24	Mon/yr.	Data	t	Regression ranges					
25	1993	4,963	1,993	X range	Y range				
26	2005	5,039	2,005	ln(t)	ln(Y)	Est ln(Y)	Est Y	e = Y - Est Y	e^2
27	2007	5,257	2,007	7.597	8.510	8.50	4927.06	35.94	1291.78
28	2017	5,432	2,017	7.603	8.525	8.55	5159.93	-120.93	14623.93
29	2026		2,026	7.604	8.567	8.56	5199.66	57.34	3288.13
30	2036		2,036	7.609	8.600	8.59	5402.32	29.68	881.04
31	2046		2,046	7.614	#N/D	8.63	5590.55	#N/D	#N/D
32				7.619	#N/D	8.67	5806.37	#N/D	#N/D
33				7.624	#N/D	8.70	6029.40	#N/D	#N/D
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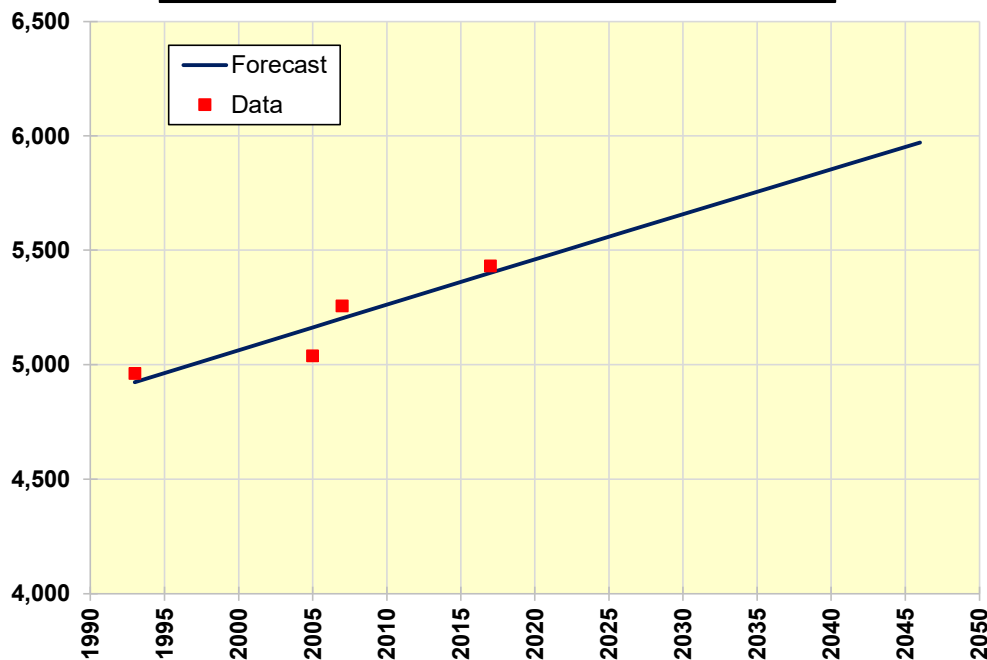
	A	B	C	D	E	F	G	H	I
1	EXPONENCIAL		IBM World-wide sales				INTERMEDIATE CALCULATIONS:		
2	Exponential regression on time						RMSE (Square root of MSE)		70.69
3	$\ln(Y_t) = a + bt$						3 X RMSE		212.08
4	Select Tools, Data Analysis, Regression. Enter ranges as follows:						Warm-up SSE		19990.10
5	Y range includes column D, X range includes column C (mismo tamaño de la columna D), Output						Warm-up MSE		4997.53
6	Ensure number of data selected for Y and X ranges agree with the						Forecasting SSE		19990.10
7	nbr. of warm-up data entered in cell C10.						Fcst SSE - Warm-up SSE		0.00
8							Nbr. of forecast periods		0
9	INPUT:						Forecasting MSE		0.00
10	Nbr. of warm-up data		4						
11	Last period to forecast		2046						
12							Warm-up Sum abs. err.		243.15
13							Warm-up MAD		60.79
14	OUTPUT:						Forecasting Sum abs. err.		243.15
15	Number of data		4				Fcst Sum abs. - Warm-up Sum at		0.00
16	Nbr. of outliers		0				Nbr. of forecast periods		0
17	Warm-up MSE		4997.53				Forecasting MAD		0.00
18	Forecasting MSE		0.00				Last forecast at period		2046
19	Warm-up MAD		60.79						
20	Forecasting MAD		0.00						
21									
22			Regression ranges						
23			X range	Y range					
24	Mon/yr.	Data	t	ln(Y)	Est ln(Y)	Est Y	e = Y - Est Y	e^2	
25	1993	4,963	1,993	8.510	8.50	4927.27	35.73	1276.91	
26	2005	5,039	2,005	8.525	8.55	5159.56	-120.56	14535.58	
27	2007	5,257	2,007	8.567	8.56	5199.33	57.67	3325.72	
28	2017	5,432	2,017	8.600	8.59	5402.81	29.19	851.89	
29	2026		2,026	#N/D	8.63	5592.75	#N/D	#N/D	
30	2036		2,036	#N/D	8.67	5811.63	#N/D	#N/D	
31	2046		2,046	#N/D	8.71	6039.07	#N/D	#N/D	
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Regresión exponencial en función del tiempo



	A	B	C	D	E	F	G	H	I
1	LOGARITMICO		IBM World-wide sales				INTERMEDIATE CALCULATIONS:		
2	Logaritmica regression on time						RMSE (Square root of MSE)		72.17
3	$Y = a + b \cdot \ln(t)$						3 X RMSE		216.50
4	Select Tools, Data Analysis, Regression. Enter ranges as follows:								
5	Y range includes column B, X range includes column D (mismo tamaño de la columna B), Output						Warm-up SSE		20832.84
6	Ensure number of data selected for Y and X ranges agree with the						Warm-up MSE		5208.21
7	nbr. of warm-up data entered in cell C10.						Forecasting SSE		20832.84
8							Fcst SSE - Warm-up SSE		0.00
9	INPUT:						Nbr. of forecast periods		0
10	Nbr. of warm-up data		4				Forecasting MSE		0.00
11	Last period to forecast		2046						
12							Warm-up Sum abs. err.		248.32
13							Warm-up MAD		62.08
14	OUTPUT:						Forecasting Sum abs. err.		248.32
15	Number of data		4				Fcst Sum abs. - Warm-up Sum at		0.00
16	Nbr. of outliers		0				Nbr. of forecast periods		0
17	Warm-up MSE		5208.21				Forecasting MAD		0.00
18	Forecasting MSE		0.00				Last forecast at period		2046
19	Warm-up MAD		62.08						
20	Forecasting MAD		0.00						
21									
22	Regression ranges								
23	<u>Y range</u>			<u>X range</u>					
24	Mon/yr.	Data	t	ln(t)	Est Y	e = Y - Est Y	e^2		
25	1993	4,963	1,993	7.597	4923.55	39.45	1556.54		
26	2005	5,039	2,005	7.603	5163.16	-124.16	15415.34		
27	2007	5,257	2,007	7.604	5202.95	54.05	2920.94		
28	2017	5,432	2,017	7.609	5401.34	30.66	940.02		
29	2026		2,026	7.614	5579.05	#N/D	#N/D		
30	2036		2,036	7.619	5775.58	#N/D	#N/D		
31	2046		2,046	7.624	5971.15	#N/D	#N/D		
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Regresión logarítmica en función del tiempo



	A	B	C	D	E	F	G	H	I
1	POLINOMICA_2		IBM World-wide sales				INTERMEDIATE CALCULATIONS:		
2	Polinómica 2do Grado regression on time						RMSE (Square root of MSE)		62.33
3	$Y = a + b*t + c*t^2$						3 X RMSE		187.00
4	Select Tools, Data Analysis, Regression. Enter ranges as follows:								
5	Y range includes column B, X range includes column C y D (mismo tamaño de la columna B), O						Warm-up SSE		15542.35
6	Ensure number of data selected for Y and X ranges agree with the						Warm-up MSE		3885.59
7	nbr. of warm-up data entered in cell C10.						Forecasting SSE		15542.35
8							Fcst SSE - Warm-up SSE		0.00
9	INPUT:						Nbr. of forecast periods		0
10	Nbr. of warm-up data		4				Forecasting MSE		0.00
11	Last period to forecast		2046						
12							Warm-up Sum abs. err.		15542.35
13							Warm-up MAD		3885.59
14	OUTPUT:						Forecasting Sum abs. err.		15542.35
15	Number of data		4				Fcst Sum abs. - Warm-up Sum at		0.00
16	Nbr. of outliers		0				Nbr. of forecast periods		0
17	Warm-up MSE		3885.59				Forecasting MAD		0.00
18	Forecasting MSE		0.00				Last forecast at period		2046
19	Warm-up MAD		3885.59						
20	Forecasting MAD		0.00						
21									
22	Regression ranges								
23	Y range		X range						
24	Mon/yr.	Data	t	t^2	Est Y	e = Y - Est Y	e^2		
25	1993	4,963	1,993	3972049	4956.82	6.18	38.25		
26	2005	5,039	2,005	4020025	5125.59	-86.59	7497.30		
27	2007	5,257	2,007	4028049	5167.94	89.06	7931.83		
28	2017	5,432	2,017	4068289	5440.66	-8.66	74.97		
29	2026		2,026	4104676	5772.97	#N/D	#N/D		
30	2036		2,036	4145296	6238.73	#N/D	#N/D		
31	2046		2,046	4186116	6806.08	#N/D	#N/D		
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Regresión polinómica_2 en función del tiempo

